Compton Scattering Cross Section in QED

Griffiths 7.30 + 7.41 Combined | H&M Gauge $\sum \varepsilon_\mu \varepsilon_\nu^* = -g_{\mu\nu}$
Mandelstam Variables | $d\sigma/dt$ | Crossing to Pair Annihilation | Virtual Photon

PHYS 7502 — Particles and Nuclei 2

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1 Problem Statement

Problem (verbatim)

Griffiths 7.30 and essentially 7.41, combined (Compton Scattering cross section) except instead of the form given in 7.41, verify the formula in Mandelstam variables given for the Matrix element in the slides from for highly relativistic particles from the Halzen & Martin textbook. Also don't use Griffiths' choice for the completeness of the epsilon photon polarization vectors but rather the more common gauge choice: $\sum(\varepsilon_\mu \varepsilon_\nu^*) = -g_{\mu\nu}$.

What This Requires

1. Write the Compton amplitude $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t$ (Griffiths Eqs. 7.111–7.113).
2. Apply Casimir's trick (Problem 7.30) to get traces.
3. Sum over photon polarizations using $-g_{\mu\nu}$ (H&M Eq. 6.111), *not* Griffiths Eq. 7.105.
4. Evaluate in the highly relativistic limit $m \to 0$.
5. Express in Mandelstam variables.
6. Verify: $\overline{|\mathcal{M}|^2} = -2e^4(t/s + s/t)$ (Griffiths convention).
7. Obtain $d\sigma/dt$.
8. Derive pair annihilation via crossing (Problem 7.41).
9. Virtual photon extension (H&M Exercise 6.19).

2 Kinematics (Griffiths Convention)

Griffiths' Particle Labeling (Example 7.4)

$$\gamma(p_3) + e^-(p_1) \longrightarrow \gamma(p_2) + e^-(p_4)$$

1:	e^- incoming	p_1	spinor $u(1)$
2:	γ **outgoing**	p_2	polarization ε_2
3:	γ **incoming**	p_3	polarization ε_3
4:	e^- outgoing	p_4	spinor $\bar{u}(4)$

Mandelstam variables:

$$s = (p_1 + p_3)^2 \tag{1}$$
$$t = (p_1 - p_2)^2 \tag{2}$$
$$u = (p_1 - p_4)^2 \tag{3}$$

$s + t + u = 2m^2$. At $m = 0$: $s + t + u = 0$, $s > 0$, $t \leq 0$, $u \leq 0$.
Dot products ($m = 0$):

$$p_1{\cdot}p_3 = \tfrac{s}{2}, \quad p_1{\cdot}p_2 = -\tfrac{t}{2}, \quad p_1{\cdot}p_4 = -\tfrac{u}{2}, \quad p_2{\cdot}p_4 = \tfrac{s}{2}, \quad p_3{\cdot}p_4 = -\tfrac{t}{2}, \quad p_2{\cdot}p_3 = -\tfrac{u}{2} \tag{4}$$

3 The Two Amplitudes (Griffiths Eqs. 7.111–7.112)

Compton Amplitudes — Griffiths Convention

s**-channel** (Fig. 7.6): e^- absorbs $\gamma_{\text{in}}(p_3)$, propagates with $q = p_1 + p_3$, emits $\gamma_{\text{out}}(p_2)$:

$$\mathcal{M}_s = \frac{-e^2}{s - m^2}\, \bar{u}(4)\, \not{\varepsilon}_2\, (\not{p}_1 + \not{p}_3 + m)\, \not{\varepsilon}_3\, u(1) \tag{5}$$

t**-channel** (Fig. 7.7): e^- emits $\gamma_{\text{out}}(p_2)$, propagates with $q = p_1 - p_2$, absorbs $\gamma_{\text{in}}(p_3)$:

$$\mathcal{M}_t = \frac{-e^2}{t - m^2}\, \bar{u}(4)\, \not{\varepsilon}_3\, (\not{p}_1 - \not{p}_2 + m)\, \not{\varepsilon}_2\, u(1) \tag{6}$$

Total: $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t$. (Add, not subtract: no fermion interchange.)

4 Photon Polarization Sum: $-g_{\mu\nu}$ (H&M Eq. 6.111)

Why $-g_{\mu\nu}$ Works

A real photon has 2 transverse polarization states. The exact completeness relation involves $k_\mu k_\nu$ terms, but the **Ward identity** ($k^\mu \mathcal{M}_\mu = 0$) guarantees these give zero when contracted with the gauge-invariant amplitude $\mathcal{M}_s + \mathcal{M}_t$. Therefore:

$$\boxed{\sum_{\lambda=1,2} \varepsilon_\mu^{(\lambda)}(k)\, \varepsilon_\nu^{(\lambda)*}(k) \;\longrightarrow\; -g_{\mu\nu}} \tag{7}$$

This is Lorentz covariant and much simpler than Griffiths' Coulomb gauge form $\delta_{ij} - \hat{k}_i \hat{k}_j$.

5 Squaring and Averaging (Problem 7.30)

We compute:

$$\overline{|\mathcal{M}|^2} \equiv \frac{1}{4} \sum_{\text{spins}} \sum_{\text{pol}} |\mathcal{M}_s + \mathcal{M}_t|^2 = \frac{1}{4} \sum \Big(|\mathcal{M}_s|^2 + |\mathcal{M}_t|^2 + \mathcal{M}_s \mathcal{M}_t^* + \mathcal{M}_s^* \mathcal{M}_t \Big) \tag{8}$$

The $\frac{1}{4}$ averages over 2 electron spins $\times$ 2 photon polarizations. These are the **four terms** noted in Problem 7.30.

Casimir's trick (Griffiths Eq. 7.125) converts each term's spin sum into a trace:

$$\sum_{\text{spins}} [\bar{u}(a)\Gamma_1 u(b)][\bar{u}(a)\Gamma_2 u(b)]^* = \text{Tr}\Big[\Gamma_1(\not{p}_b + m)\bar{\Gamma}_2(\not{p}_a + m)\Big]$$

At $m = 0$: $\not{p} + m \to \not{p}$.

6 Evaluating $\frac{1}{4}\sum |\mathcal{M}_s|^2$

With $m = 0$, the s-channel gives after Casimir + photon sums ($-g_{\mu\alpha}$ and $-g_{\nu\beta}$):

$$\frac{1}{4} \sum |\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\, \text{Tr}\Big[\gamma^\mu(\not{p}_1 + \not{p}_3)\gamma^\nu\, \not{p}_1\, \gamma_\nu(\not{p}_1 + \not{p}_3)\gamma_\mu\, \not{p}_4\Big] \tag{9}$$

Step 1: Inner contraction

$\gamma^\nu \not{p}_1 \gamma_\nu = -2\not{p}_1$ (Griffiths Thm. 7).

Step 2: Central product

$\not{p}_1^2 = p_1^2 = 0$ kills the $\not{p}_1\not{p}_1$ terms:

$$(\not{p}_1 + \not{p}_3)\not{p}_1(\not{p}_1 + \not{p}_3) = \not{p}_3\not{p}_1\not{p}_3 \tag{10}$$

Using $\not{p}_3\not{p}_1 = 2p_3\cdot p_1 - \not{p}_1\not{p}_3$ and $\not{p}_3^2 = 0$:

$$\not{p}_3\not{p}_1\not{p}_3 = 2(p_1\cdot p_3)\not{p}_3 = s\,\not{p}_3$$

Key Identity (s-channel)

$$(\not{p}_1 + \not{p}_3)\not{p}_1(\not{p}_1 + \not{p}_3) = s\,\not{p}_3$$

Step 3: Outer contraction

$\gamma^\mu \not{p}_3 \gamma_\mu = -2\not{p}_3$. Collecting:

$$\frac{e^4}{4s^2}(-2)(s)(-2)\,\mathrm{Tr}(\not{p}_3\not{p}_4) = \frac{e^4}{s}\,\mathrm{Tr}(\not{p}_3\not{p}_4)$$

Step 4: Trace

$\mathrm{Tr}(\not{p}_3\not{p}_4) = 4\,p_3{\cdot}p_4 = 4(-t/2) = -2t$.

s-Channel Result

$$\boxed{\frac{1}{4}\sum |\mathcal{M}_s|^2 = -\frac{2e^4 t}{s}} \tag{11}$$

7 Evaluating $\frac{1}{4}\sum |\mathcal{M}_t|^2$

Same structure with $p_3 \to -p_2$, denominator $s \to t$:

$$\frac{1}{4}\sum |\mathcal{M}_t|^2 = \frac{e^4}{4t^2}\,\mathrm{Tr}\Big[\gamma^\mu(\not{p}_1 - \not{p}_2)\gamma^\nu\, \not{p}_1\, \gamma_\nu(\not{p}_1 - \not{p}_2)\gamma_\mu\, \not{p}_4\Big] \tag{12}$$

Steps 1–2

Inner contraction: $\gamma^\nu \not{p}_1 \gamma_\nu = -2\not{p}_1$. Central product ($\not{p}_1^2 = 0$):

$$(\not{p}_1 - \not{p}_2)\not{p}_1(\not{p}_1 - \not{p}_2) = \not{p}_2\not{p}_1\not{p}_2 = 2(p_1{\cdot}p_2)\not{p}_2 = (-t)\not{p}_2$$

Key Identity (t-channel)

$$(\not{p}_1 - \not{p}_2)\not{p}_1(\not{p}_1 - \not{p}_2) = (-t)\,\not{p}_2$$

Steps 3–4

Outer: $\gamma^\mu \not{p}_2 \gamma_\mu = -2\not{p}_2$. Trace: $\mathrm{Tr}(\not{p}_2\not{p}_4) = 4\,p_2{\cdot}p_4 = 2s$.

Collecting:

$$\frac{e^4}{4t^2}(-2)(-t)(-2)(2s) = -\frac{2e^4 s}{t}$$

t-Channel Result

$$\boxed{\frac{1}{4}\sum |\mathcal{M}_t|^2 = -\frac{2e^4 s}{t}} \tag{13}$$

8 Interference Terms Vanish at $m = 0$

Why $\mathcal{M}_s^* \mathcal{M}_t + \text{c.c.} = 0$ at $m = 0$

Physical: At $m = 0$, helicity is conserved at each QED vertex. The s- and t-channels connect different intermediate helicity configurations, so their interference vanishes.
Technical: After $-g_{\mu\nu}$ and Casimir, the interference traces produce terms that vanish when all external particles are massless.

9 Main Result: $\overline{|\mathcal{M}|^2}$

Combining (11) and (13):

Compton Scattering $\overline{|\mathcal{M}|^2}$ (Griffiths Convention, $m = 0$)

$$\boxed{\overline{|\mathcal{M}|^2} = -2e^4\left(\frac{t}{s} + \frac{s}{t}\right) = -2e^4\,\frac{s^2+t^2}{st}} \tag{14}$$

Positive check: $s > 0$, $t < 0 \Rightarrow -t/s > 0$, $-s/t > 0 \Rightarrow \overline{|\mathcal{M}|^2} > 0$. ✓

This is the H&M result (Eq. post-6.112, with H&M's s, u corresponding to Griffiths' s, t).

Convention translation: H&M/L&B/P&S write $-2e^4(u/s+s/u)$ because their "u" is Griffiths' "t" (they label outgoing particles 3,4 instead of 2,4).

10 Invariant Cross Section: $d\sigma/dt$

Thomson Eq. 3.37

For $2 \to 2$ scattering:

$$\frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{64\pi\, s\, p_i^{*2}}$$

At $m = 0$: $p_i^{*2} = s/4$, so $\dfrac{d\sigma}{dt} = \dfrac{\overline{|\mathcal{M}|^2}}{16\pi\, s^2}$.

Using $e^4 = 16\pi^2\alpha^2$:

Compton $d\sigma/dt$

$$\boxed{\frac{d\sigma}{dt} = -\frac{2\pi\alpha^2(s^2+t^2)}{s^3\, t} = \frac{2\pi\alpha^2(s^2+t^2)}{s^3|t|}} \tag{15}$$

Using $u = -s - t$: $\dfrac{d\sigma}{dt} = \dfrac{2\pi\alpha^2[s^2 + (s+t)^2]}{s^3(s+t)}$

11 Crossing Symmetry: Pair Annihilation (Problem 7.41)

Crossing: Compton $\to e^+e^- \to \gamma\gamma$

Cross the incoming photon to outgoing, and the outgoing electron to an incoming positron:

$$p_1 \to q_1, \quad p_2 \to q_4, \quad p_3 \to -q_3, \quad p_4 \to -q_2$$

This maps Mandelstam variables cyclically:

$$\boxed{s_C \to t_A, \qquad t_C \to u_A, \qquad u_C \to s_A} \tag{16}$$

The s-channel (Compton) becomes the t-channel (annihilation).

Substituting $s \to t$, $t \to u$ into (14):

Pair Annihilation $e^+e^- \to \gamma\gamma$ ($m = 0$)

$$\boxed{\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} = -2e^4\left(\frac{u}{t} + \frac{t}{u}\right)} \tag{17}$$

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{e^+e^-\to\gamma\gamma} = -\frac{2\pi\alpha^2(t^2+u^2)}{s^2\,tu}} \tag{18}$$

12 Virtual Photon Extension (H&M Exercise 6.19)

For $k^2 = p_3^2 = -Q^2 \neq 0$, the key identity changes because $\not{p}_3^2 = -Q^2 \neq 0$:

$$\not{p}_3\not{p}_1\not{p}_3 = 2(p_1\cdot p_3)\not{p}_3 - \not{p}_1 \underbrace{\not{p}_3^2}_{=-Q^2} = (s+Q^2)\not{p}_3 + Q^2\not{p}_1$$

The extra $Q^2\not{p}_1$ generates corrections in both channels and makes the interference nonzero:

Virtual Photon Compton ($\gamma^* e^- \to \gamma e^-$)

$$\boxed{\overline{|\mathcal{M}|^2} = 2e^4\left(-\frac{t}{s} - \frac{s}{t} + \frac{2Q^2u}{st}\right)} \tag{19}$$

Setting $Q^2 = 0$ recovers the real Compton result. ✓

13 Helicity and Wigner d-Functions

In the CM frame at $m = 0$, the angular dependence is:

$$-\frac{t}{s} = \frac{1-\cos\theta^*}{2} = \sin^2\frac{\theta^*}{2}$$

This equals the Wigner d-function $d^1_{1,-1}(\theta^*)$ for helicity-1 rotation, confirming the result independently via angular momentum conservation.

14 Summary

Complete Roadmap

1. **Amplitudes:** $\mathcal{M}_s$ (Eq. 7.111) + $\mathcal{M}_t$ (Eq. 7.112), propagators $1/s$ and $1/t$.
2. $|\mathcal{M}|^2$**:** 4 terms (Problem 7.30).
3. **Casimir:** $\sum u\bar{u} = \not{p}$ converts spins to traces.
4. **Photon sums:** $\sum \varepsilon_\mu \varepsilon_\nu^* \to -g_{\mu\nu}$ (H&M 6.111), justified by Ward identity.
5. **Key identities:** $(\not{p}_1 + \not{p}_3)\not{p}_1(\not{p}_1 + \not{p}_3) = s\,\not{p}_3$ and $(\not{p}_1 - \not{p}_2)\not{p}_1(\not{p}_1 - \not{p}_2) = (-t)\,\not{p}_2$.
6. **Traces:** $\mathrm{Tr}(\not{p}_3\not{p}_4) = -2t$, $\mathrm{Tr}(\not{p}_2\not{p}_4) = 2s$.
7. **Result:** $\overline{|\mathcal{M}|^2} = -2e^4(t/s + s/t)$.
8. **Cross section:** $d\sigma/dt = -2\pi\alpha^2(s^2+t^2)/(s^3 t)$.
9. **Crossing** ($s \to t$, $t \to u$): $\overline{|\mathcal{M}|^2}_{\text{ann}} = -2e^4(u/t + t/u)$.
10. **Virtual:** $k^2 = -Q^2$ adds $+2Q^2 u/(st)$.

Appendix: Identities

Quick Reference

$\gamma^\mu\gamma_\mu = 4$. $\quad \gamma^\mu \not{a}\gamma_\mu = -2\not{a}$. $\quad \gamma^\mu \not{a}\not{b}\gamma_\mu = 4\,a\cdot b$. $\quad \gamma^\mu \not{a}\not{b}\not{c}\gamma_\mu = -2\not{c}\not{b}\not{a}$.

$\mathrm{Tr}(\mathbf{1}) = 4$. $\quad \mathrm{Tr}(\text{odd }\#) = 0$. $\quad \mathrm{Tr}(\not{a}\not{b}) = 4\,a\cdot b$. $\quad \mathrm{Tr}(\not{a}\not{b}\not{c}\not{d}) = 4[(a\cdot b)(c\cdot d) - (a\cdot c)(b\cdot d) + (a\cdot d)(b\cdot c)]$.

$\sum u\bar{u} = \not{p} + m$. $\quad \sum v\bar{v} = \not{p} - m$. $\quad \sum \varepsilon_\mu \varepsilon_\nu^* \to -g_{\mu\nu}$.